

SOFTWARE ENGINEERING LAB – 2

(UE24CS341A)

Agile Backlog Creation & Sprint Simulation in Jira

Team Number: 003

Individual Submission: Mallikarjuna Rao R V (PES1UG24AM155)

Problem Statement 3: Campus Placement & Internship Pipeline

EPIC: Epic 3 – Interview Rounds & Status Tracking (FR-003 and FR-005)

Note: This report covers only Epic 3, my individual contribution from the Lab 1 report. The remaining requirements are covered by my teammates (Epic 1 – FR-001: Mounik; Epic 2 – FR-002 and NFR-002: Kshitij; Epic 4 – FR-004 and NFR-001: Moksha).

Lab Objective

This lab converts requirements FR-003 and FR-005 from the Lab 1 report into an Agile Epic and User Stories in Jira, estimates effort with Fibonacci story points, prioritises the backlog, simulates two one-week sprints on a Scrum board, and tracks progress using the Burndown Chart.

1. Epic (Derived from FR-003 and FR-005)

FR-003 (multi-round interview queues) and FR-005 (real-time status for students) both concern how an applicant moves through the interview process, so they are grouped into one Epic.

Epic Key	Epic Name	Mapped Requirements	Description
SCRUM-7	Epic 3: Interview Rounds & Status Tracking	FR-003, FR-005	Enable Placement Officers to manage multi-round interview queues (Aptitude, Technical, HR) and give Student Applicants real-time visibility of their application and interview-round status.

2. User Stories for Epic 3

The Epic is broken into 2 User Stories using the format: As a [role], I want [goal], so that [benefit].

Key	Story	As a...	I want...	So that...	Covers
SCRUM-8	3.1 Manage Multi-Round Interviews	Placement Officer	to create and manage multi-round interview queues and move eligible applicants through Aptitude, Technical and HR rounds	candidates can progress through the placement process in the correct sequence	FR-003
SCRUM-9	3.2 Track Application & Interview Status	Student Applicant	to view my application and interview-round status on my dashboard	I can track my progress through the placement process without contacting the Placement Officer	FR-005

Acceptance criteria were taken from Lab 1. FR-003: an applicant who clears Round 1 is automatically queued for Round 2 within 5 minutes. FR-005: the status on the dashboard matches the status last updated by the Placement Officer.

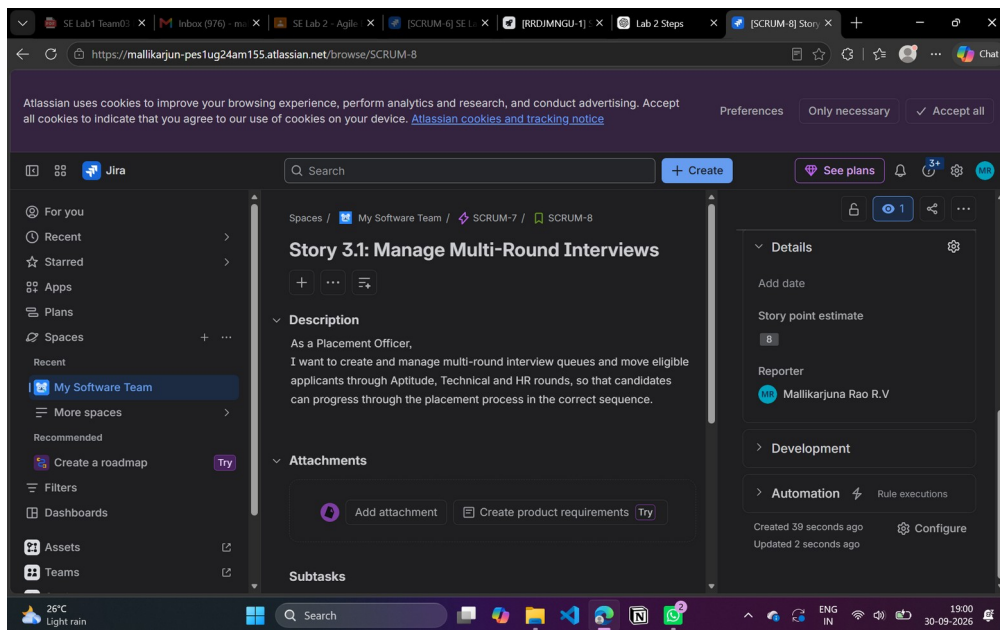


Figure 1: Story 3.1 (SCRUM-8) under Epic SCRUM-7 – description and 8 story points.

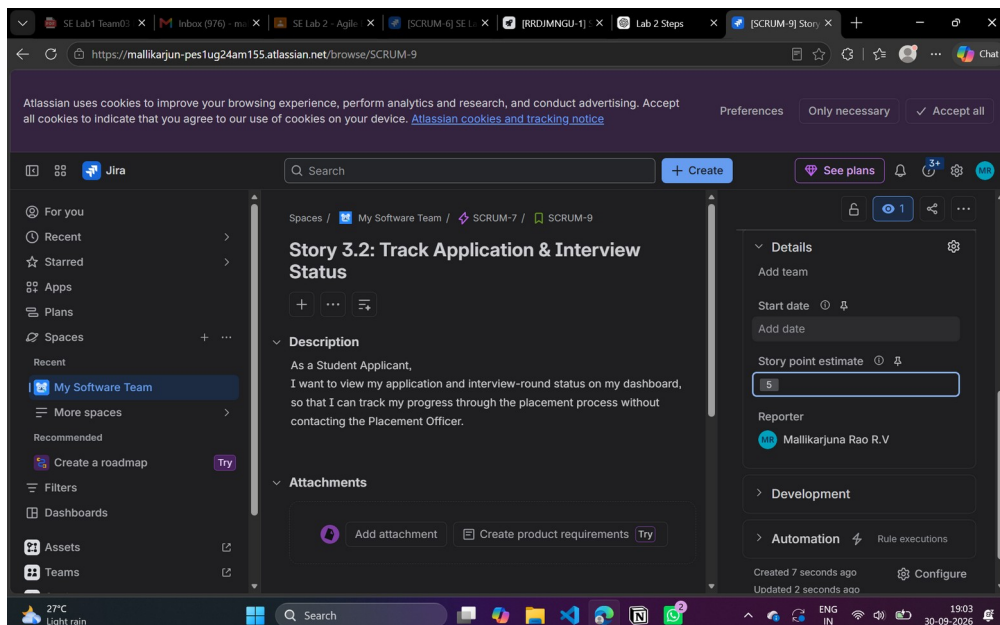


Figure 2: Story 3.2 (SCRUM-9) under Epic SCRUM-7 – description and 5 story points.

3. Jira Backlog – Epic, User Stories and Story Points

Both stories belong to Epic 3 and carry story points on the Fibonacci scale (8 and 5, 13 points in total). Both were moved into SCRUM Sprint 1 (30 Sep – 7 Oct), which shows an estimate of 13 of 13.

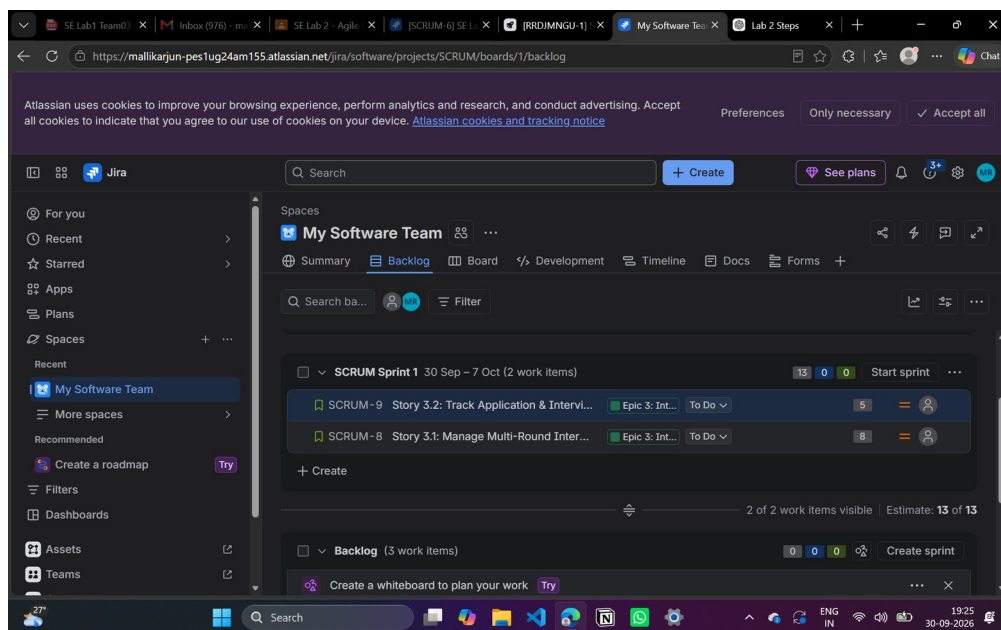


Figure 3: Jira Backlog – SCRUM Sprint 1 with SCRUM-9 (5 pts) and SCRUM-8 (8 pts) under Epic 3.

4. Backlog Prioritization

Key	Story	Priority	Story Points
SCRUM-9	3.2 Track Application & Interview Status	Medium	5
SCRUM-8	3.1 Manage Multi-Round Interviews	Medium	8

- Both stories are Medium priority, matching the priorities of FR-003 and FR-005 in Lab 1. They support the core pipeline (FR-001, FR-002, FR-004) rather than being blocking items for it.
- Story 3.2 (5 pts) was placed first in the sprint as the smaller story that gives students visibility early; Story 3.1 (8 pts) is the larger, more complex story covering round queues and sequential movement.

Total Story Points (Epic 3): 13 | High: 0 | Medium: 13

5. Story Point Estimation – Planning Poker

Story points were assigned on the Fibonacci scale (1, 2, 3, 5, 8, 13...) using the planning-poker approach, based on complexity, effort and uncertainty. The agreed estimates are below.

Story ID	Story Title	Agreed Points	Reasoning
SCRUM-8 / 3.1	Manage Multi-Round Interviews	8	Multiple rounds, queue management and automatic movement between rounds
SCRUM-9 / 3.2	Track Application & Interview Status	5	Dashboard display, depends on status updates from the officer

6. Sprint Planning & Simulation

Sprint 1 (Duration: 1 week, 30 Sep – 7 Oct)

Sprint Goal: Complete interview round management and application status tracking.

Key	Story	Priority	Pts	Status
SCRUM-9	3.2 Track Application & Interview Status	Medium	5	Done
SCRUM-8	3.1 Manage Multi-Round Interviews	Medium	8	Done

Sprint 1 Velocity: 13 pts committed | 13 pts completed | 0 pts rolled over

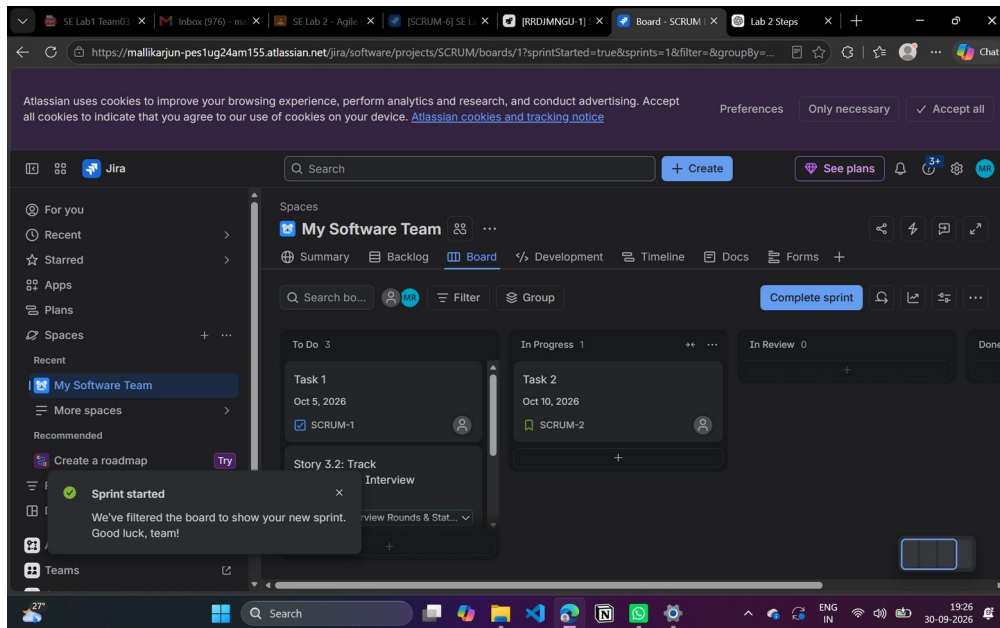


Figure 4: Sprint 1 board right after starting the sprint – Story 3.1 and Story 3.2 in To Do.

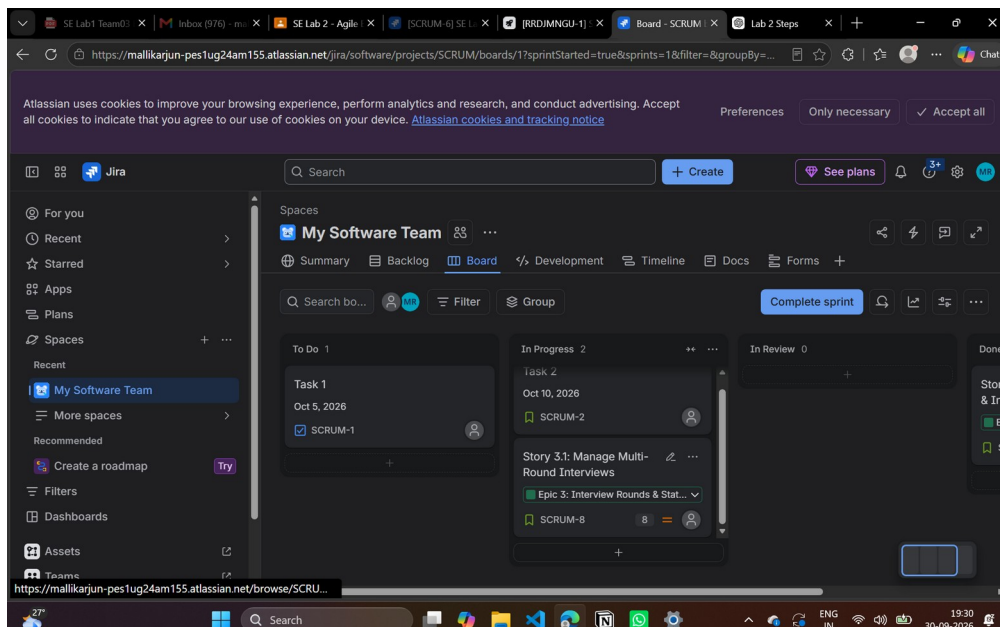


Figure 5: Sprint 1 board mid-sprint – Story 3.2 in Done; Story 3.1 In Progress.

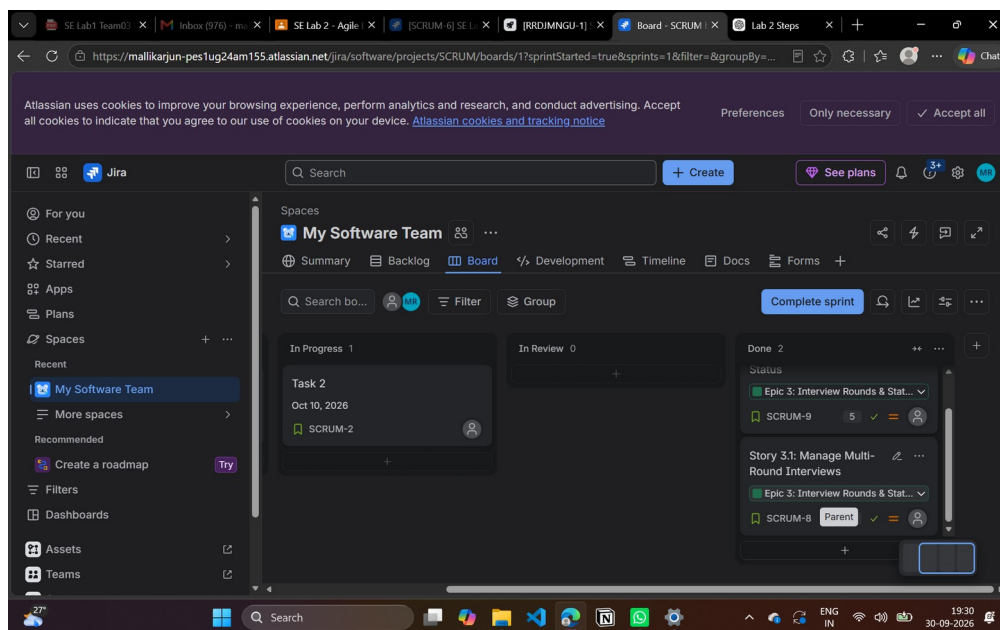


Figure 6: Sprint 1 board – both stories (13 pts) in Done, before clicking Complete sprint.

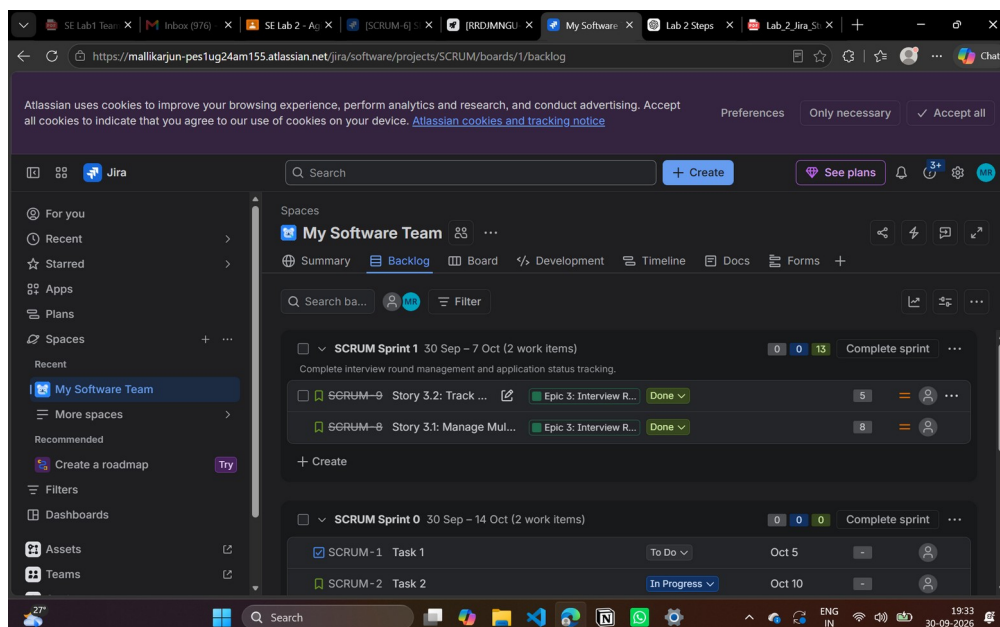


Figure 7: Backlog view after completing the stories – SCRUM-9 and SCRUM-8 marked Done in Sprint 1.

Sprint 2 (Duration: 1 week, 30 Sep – 7 Oct)

Sprint Goal: Complete the SE Lab-2 setup and organize the remaining work items for the next phase of development.

Both stories of Epic 3 were completed in Sprint 1, so Sprint 2 was used for the two remaining work items in the project, SCRUM-5 and SCRUM-6 (SE Lab-2 setup). These items were not story-pointed, so Sprint 2 is tracked by work item count instead of story points.

Key	Work item	Priority	Estimate	Status
SCRUM-5	SE Lab-2	Medium	—	Done
SCRUM-6	SE Lab-2	Medium	—	Done

Sprint 2 Velocity: 2 work items committed | 2 completed | 0 rolled over

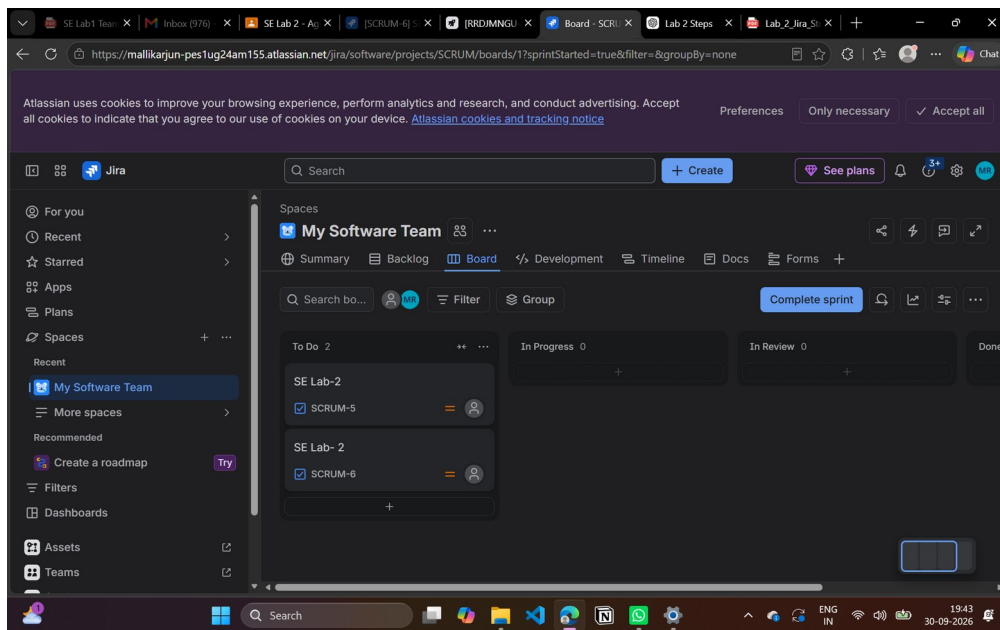


Figure 8: Sprint 2 board – SCRUM-5 and SCRUM-6 in To Do.

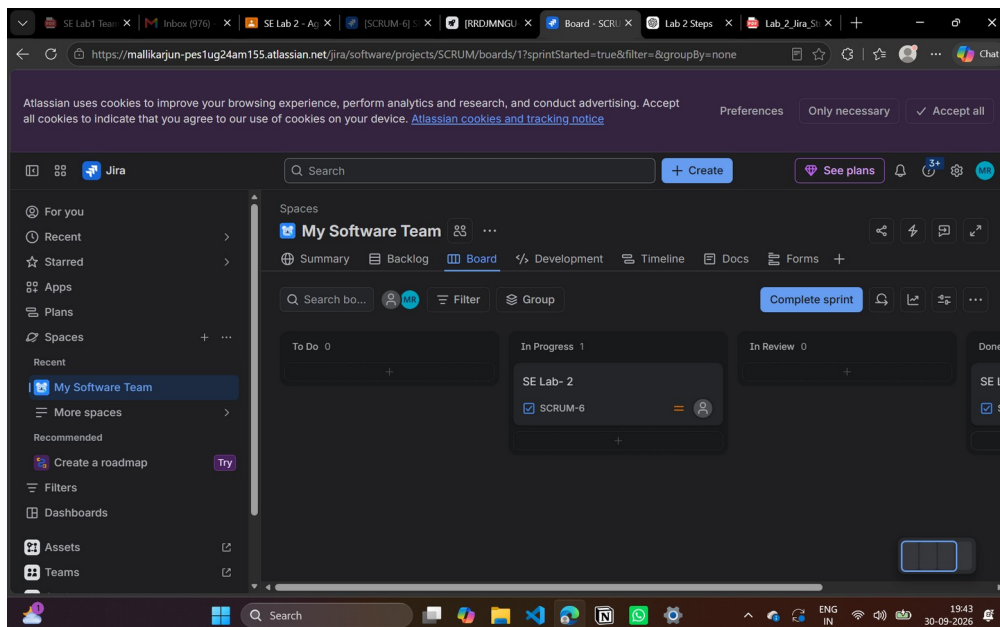


Figure 9: Sprint 2 board mid-sprint – SCRUM-5 in Done; SCRUM-6 In Progress.

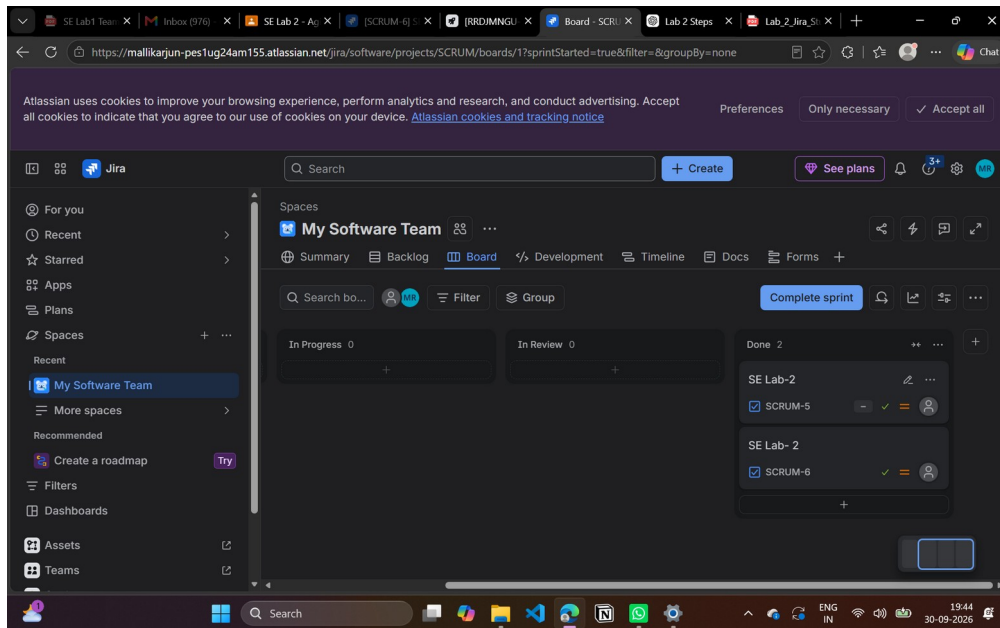


Figure 10: Sprint 2 board – both work items in Done, before clicking Complete sprint.

7. Burndown Chart Analysis

The Burndown Chart plots the work remaining (red line) against the ideal guideline (grey line).

Sprint 1 Burndown (13 pts)

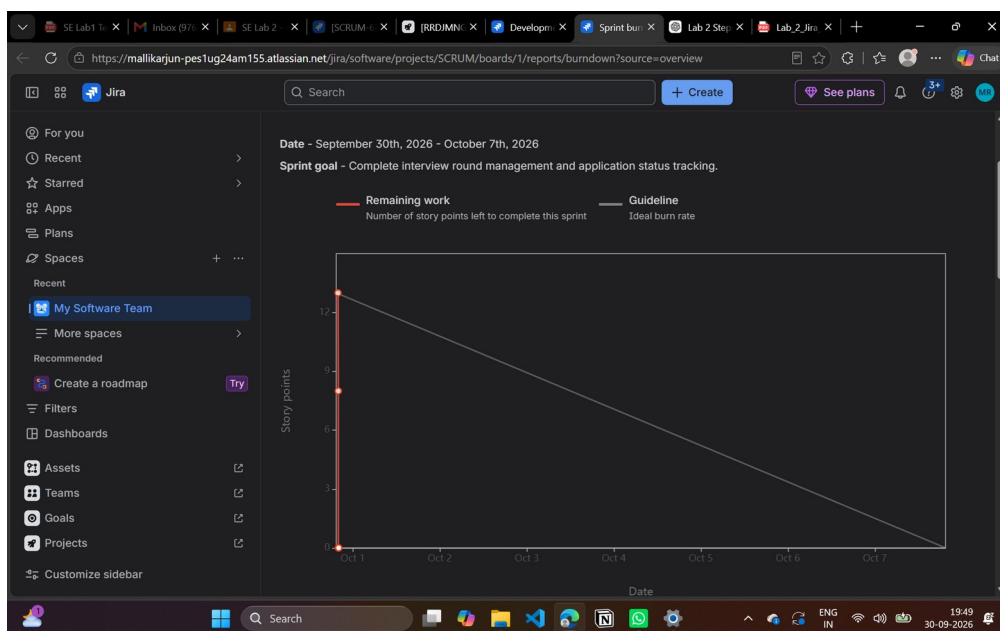


Figure 11: Sprint 1 Burndown – 13 → 8 → 0 story points on 30 Sep (Story 3.2 completed first, then Story 3.1).

Sprint 2 Burndown (2 work items)

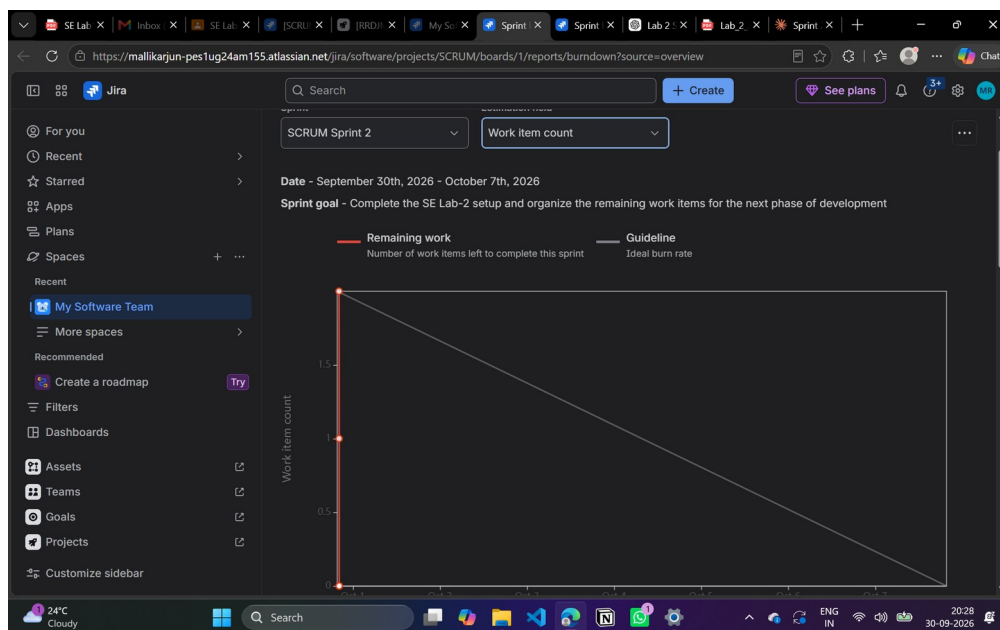


Figure 12: Sprint 2 Burndown (work item count) – 2 → 1 → 0 on 30 Sep.

Sprint	Start	Completion events	End
Sprint 1	13 points	SCRUM-9 (-5) → 8; SCRUM-8 (-8) → 0	0 points
Sprint 2	2 work items	SCRUM-5 (-1) → 1; SCRUM-6 (-1) → 0	0 work items

8. Reflection Questions

1. Did your estimations reflect the actual effort?

Only partly, since the sprints were simulated by moving cards on the board and no real development was done, so the estimates cannot be checked against actual hours. What the exercise did show is that the relative sizing was sensible: Story 3.1 (8 points) involves several interview rounds and automatic movement of applicants between them, so it carries more complexity and uncertainty than Story 3.2 (5 points), which mainly displays status. In a real sprint, 3.1 would be the story most likely to overrun.

2. Was your backlog well-prioritized?

Mostly. Both stories are Medium priority because they build on the eligibility and drive configuration work (FR-001, FR-002) and fit together in one sprint. One improvement: Story 3.1 should logically come before Story 3.2, because the status students see depends on applicants moving through rounds. I placed the smaller Story 3.2 first, so ordering the stories by dependency would have been better.

3. How did your simulated sprint align with your plan?

Sprint 1 went as planned: 13 points committed and 13 completed with nothing rolled over, and the sprint goal was met. Stories moved from To Do to In Progress to Done one after the other. Sprint 2 also completed both of its items. Because the work was simulated, each sprint finished within minutes instead of the planned week, so the 100% completion reflects the simulation, not real delivery capacity.

4. What insights did the burndown chart give about your team's capacity?

In both sprints the red line fell to zero within minutes, far below the grey guideline, which appears as a near-vertical line at the left edge of the chart. In a real sprint this would suggest under-commitment. Sprint 1 burned 13 points in two steps, and the 8-point Story 3.1 alone accounted for more than half of it in one step, which suggests large stories should be split so progress is visible throughout the sprint. A realistic velocity would need to be re-measured with real work.